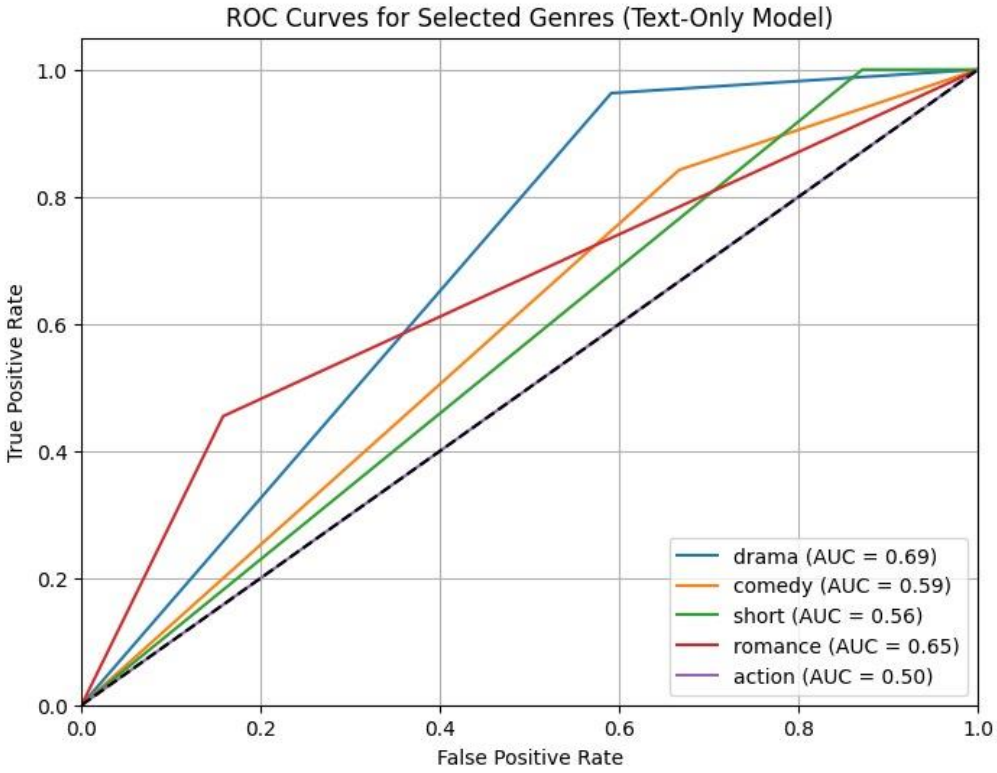
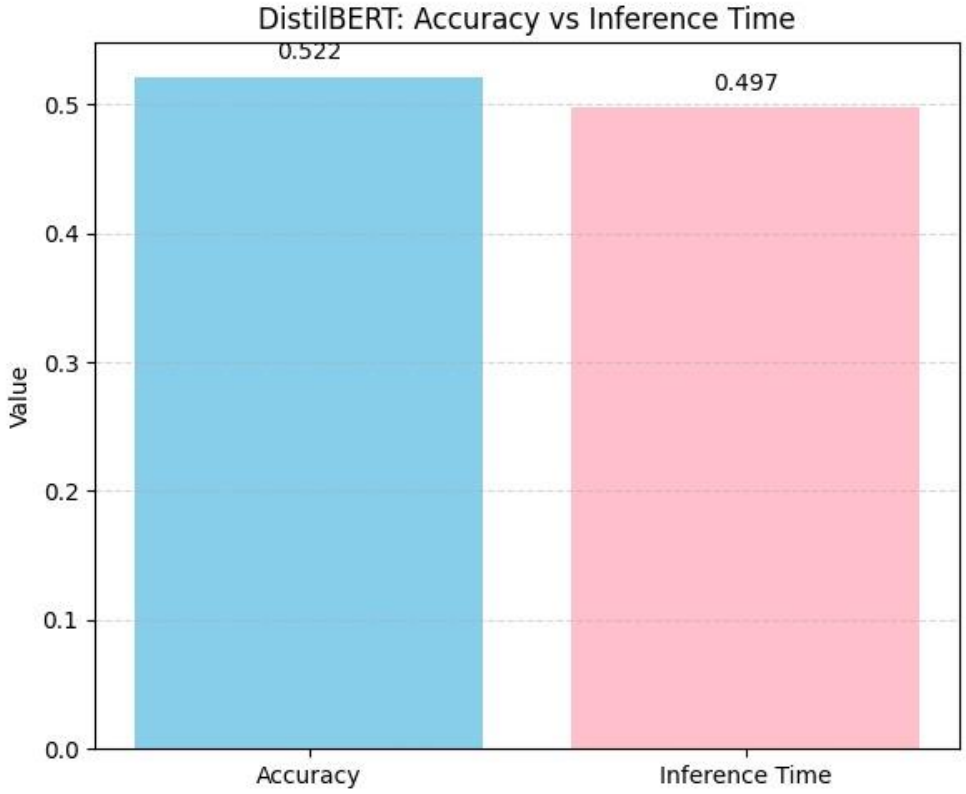
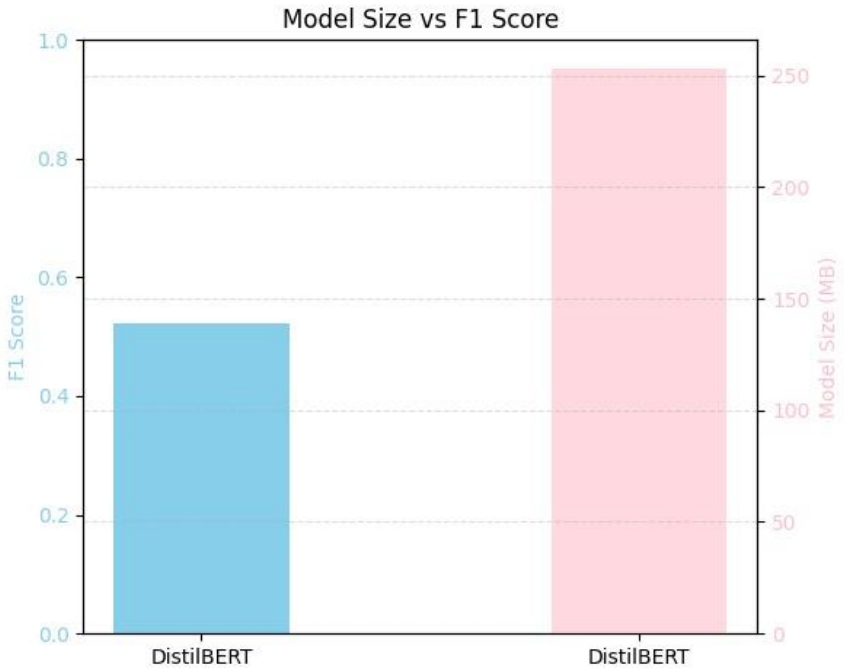
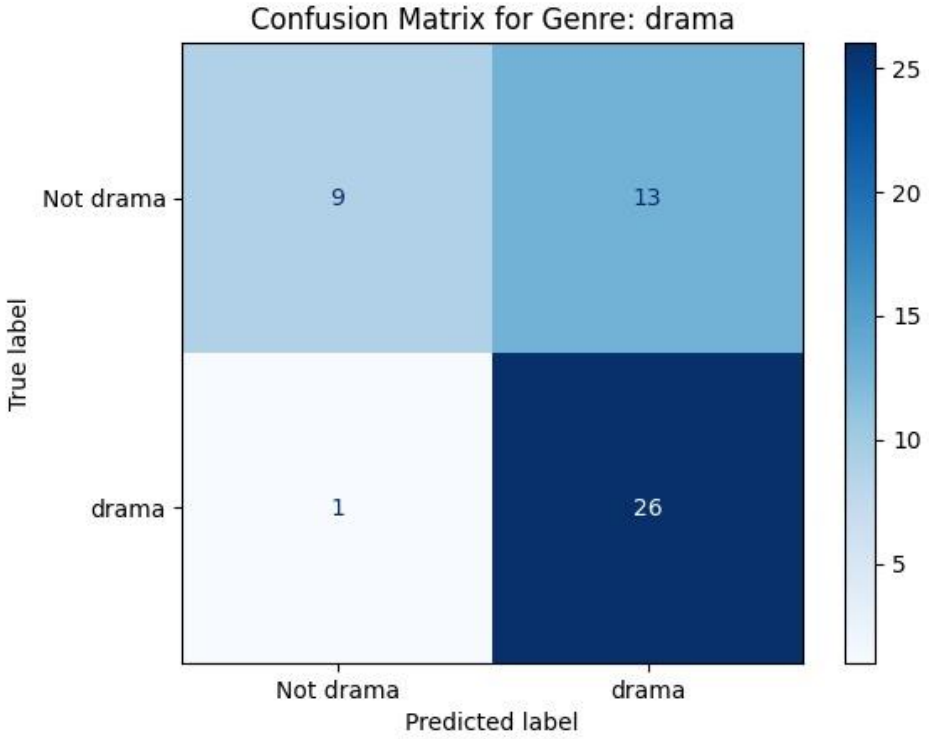
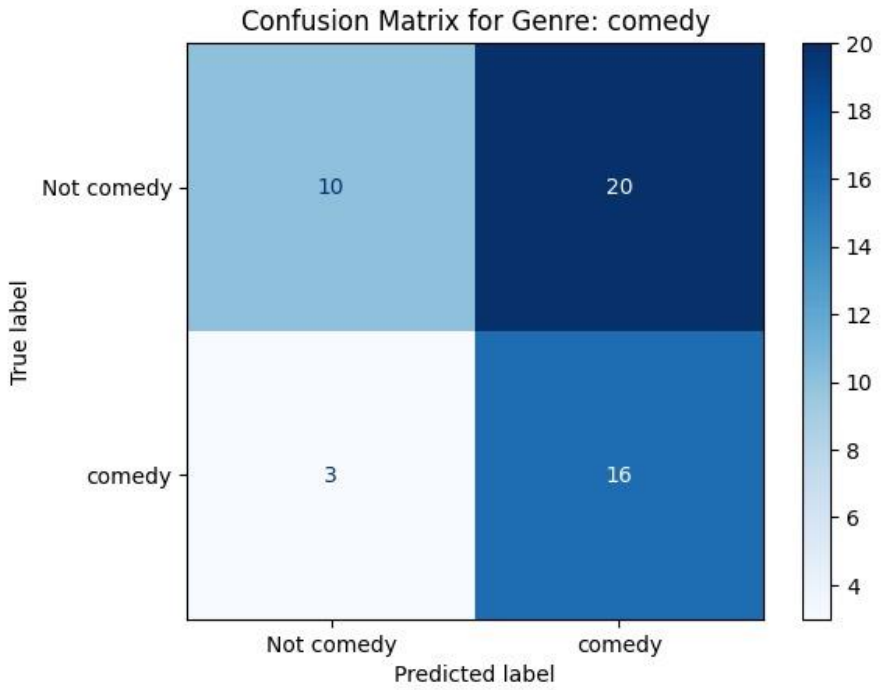
Text only unimodel (MM-imdb)

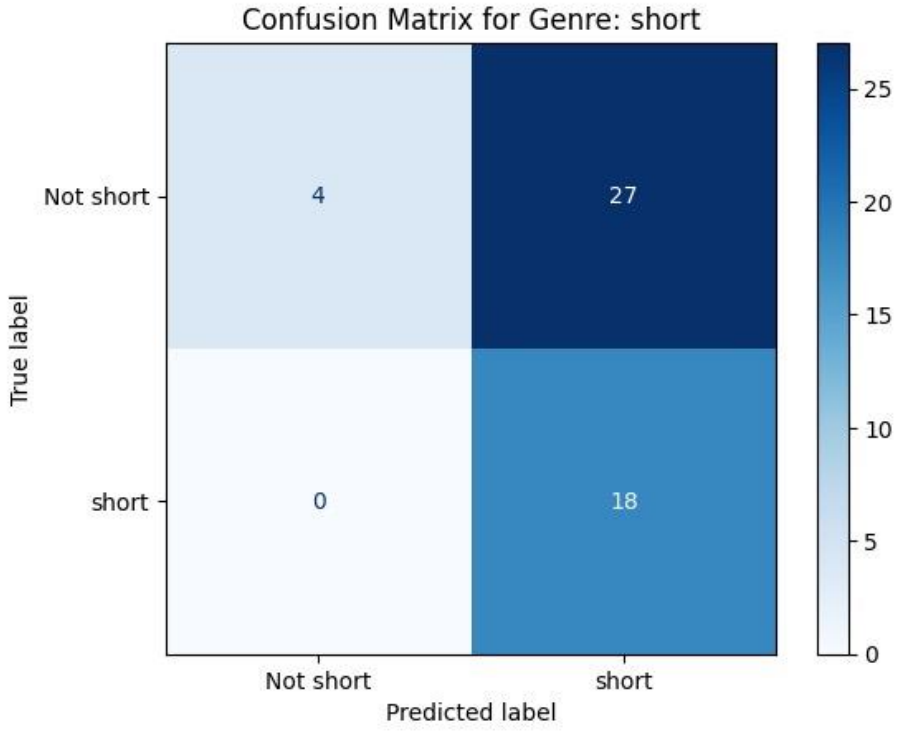


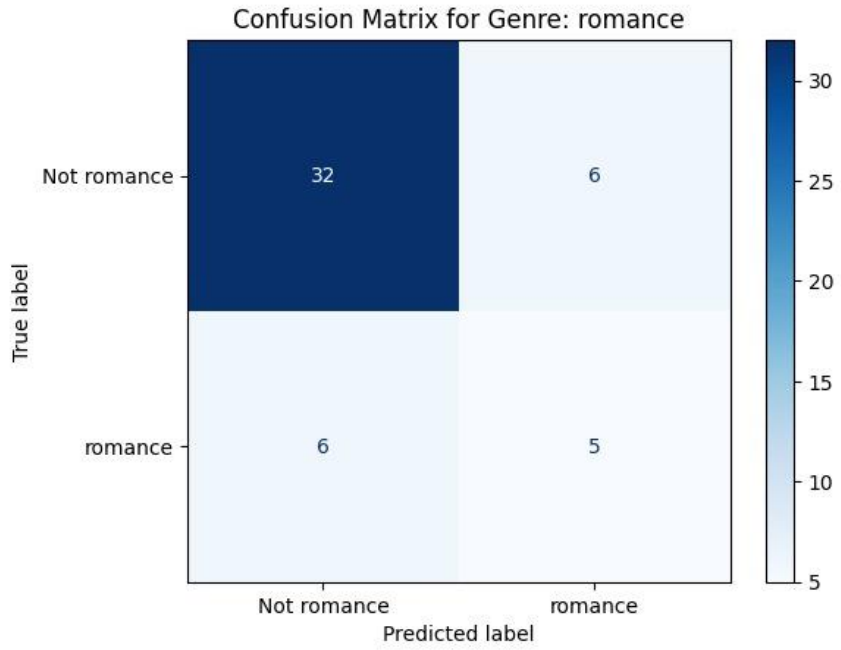


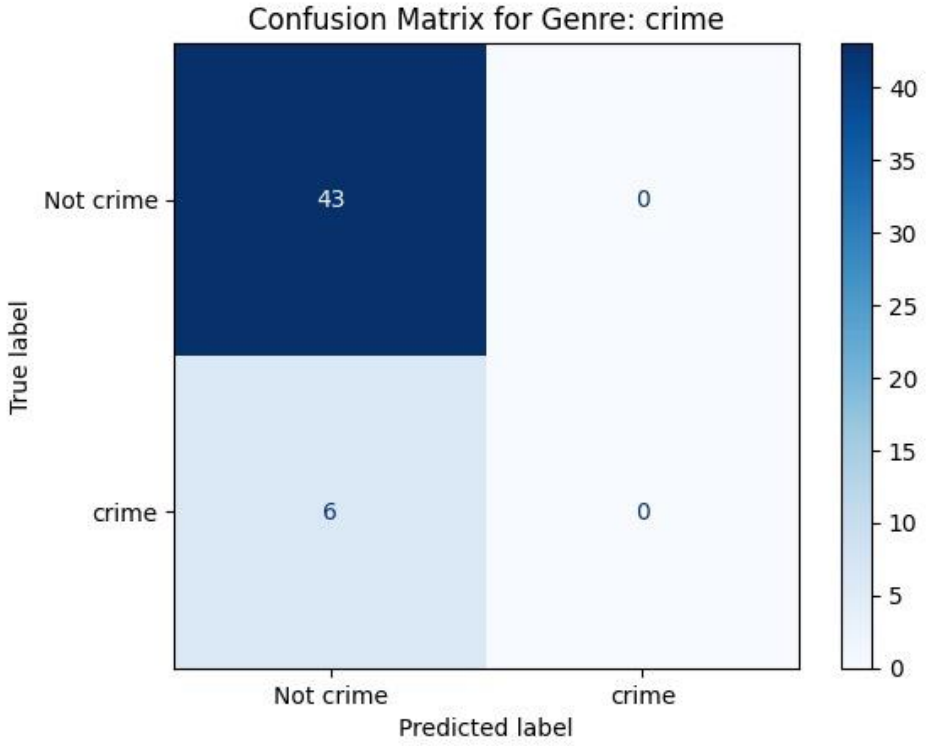

Model Size vs F1 Score
F1 Score
Model Size (MB)
1.0
0.8
0.6
0.4
0.2
0.0
250
200
150
100
50
0
DistilBERT
DistilBERT


Confusion Matrix for Genre: drama
True label
Not drama
drama
9
13
1
26
Not drama
drama
Predicted label
25
20
15
10
5

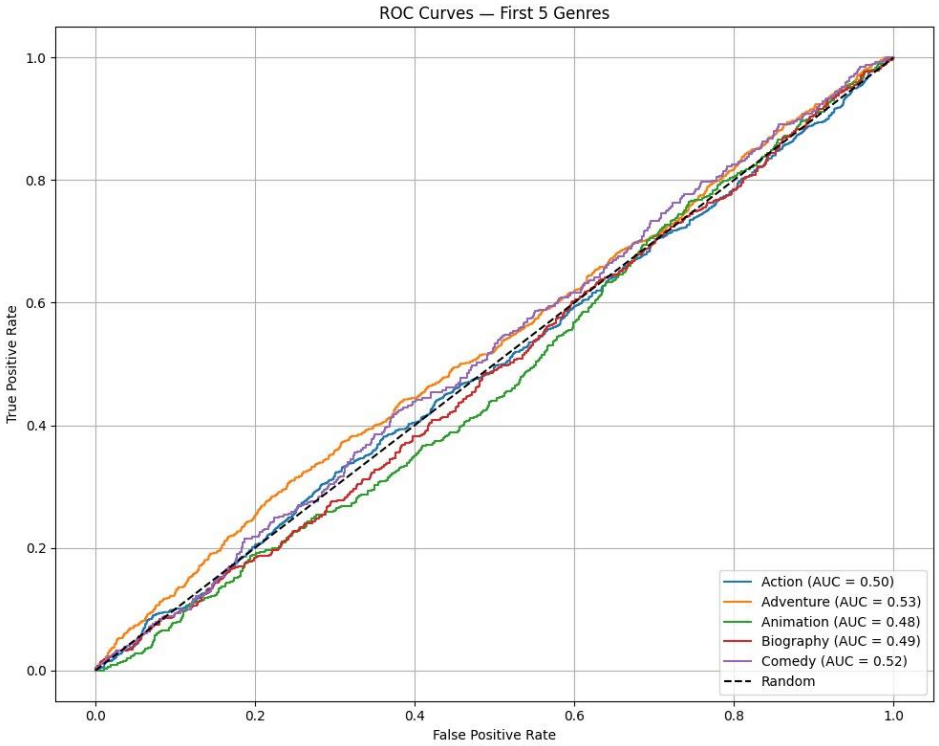

Confusion Matrix for Genre: comedy
Not comedy
comedy
True label
10
20
3
16
Not comedy
comedy
Predicted label
20
18
16
14
12
10
8
6
4

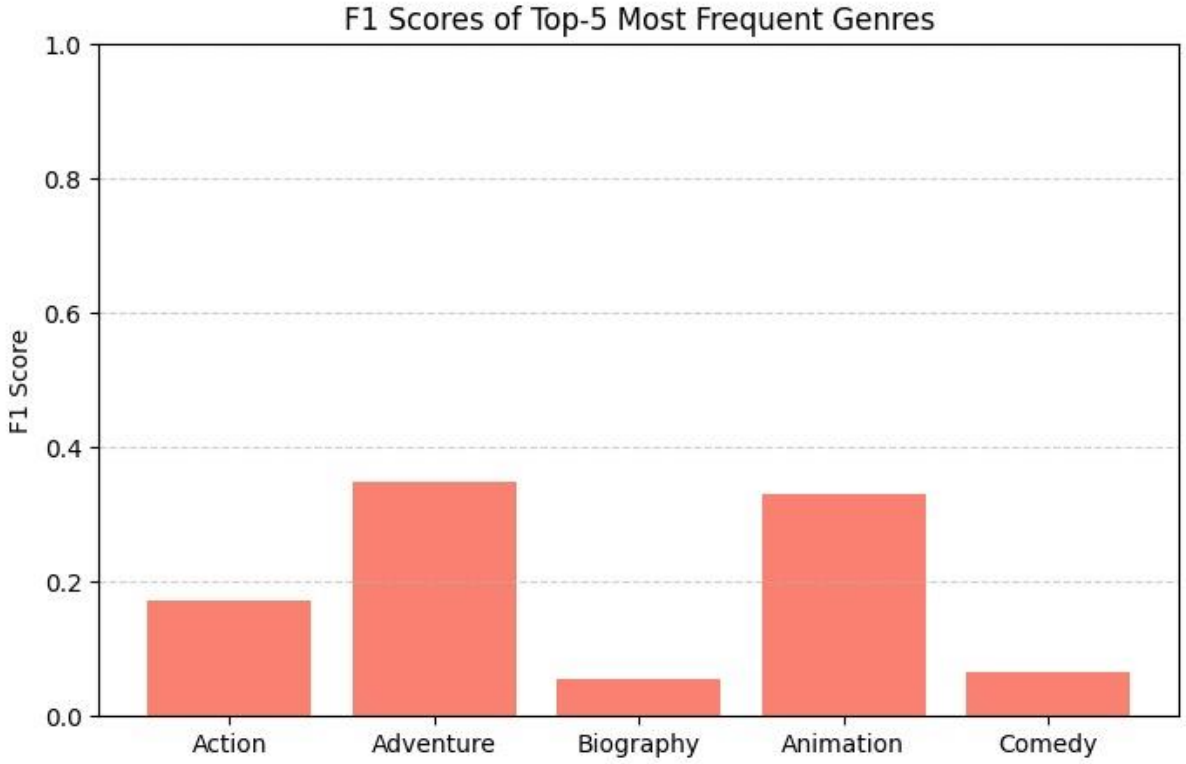

Confusion Matrix for Genre: short
Not short
short
True label
4
27
0
18
Not short
short
Predicted label
25
20
15
10
5
0

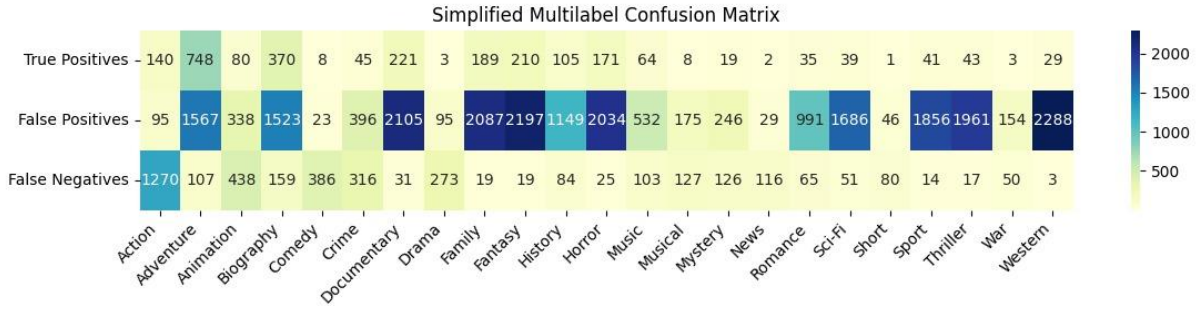

Confusion Matrix for Genre: romance
Not romance
romance
True label
32
6
6
5
Not romance
romance
Predicted label
30
25
20
15
10
5

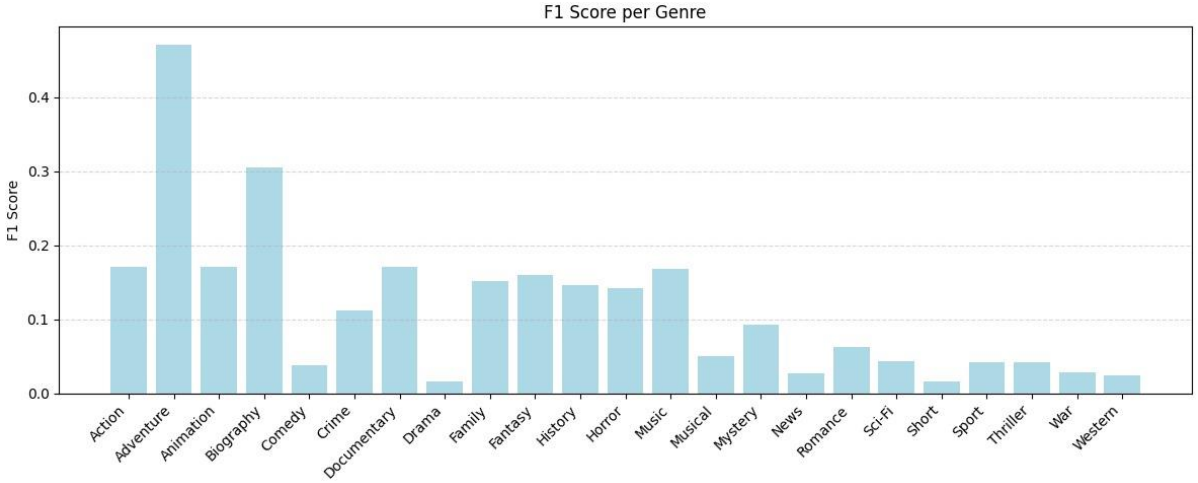

Confusion Matrix for Genre: crime
Not crime
crime
True label
43
0
6
0
Not crime
crime
Predicted label
40
35
30
25
20
15
10
5
0

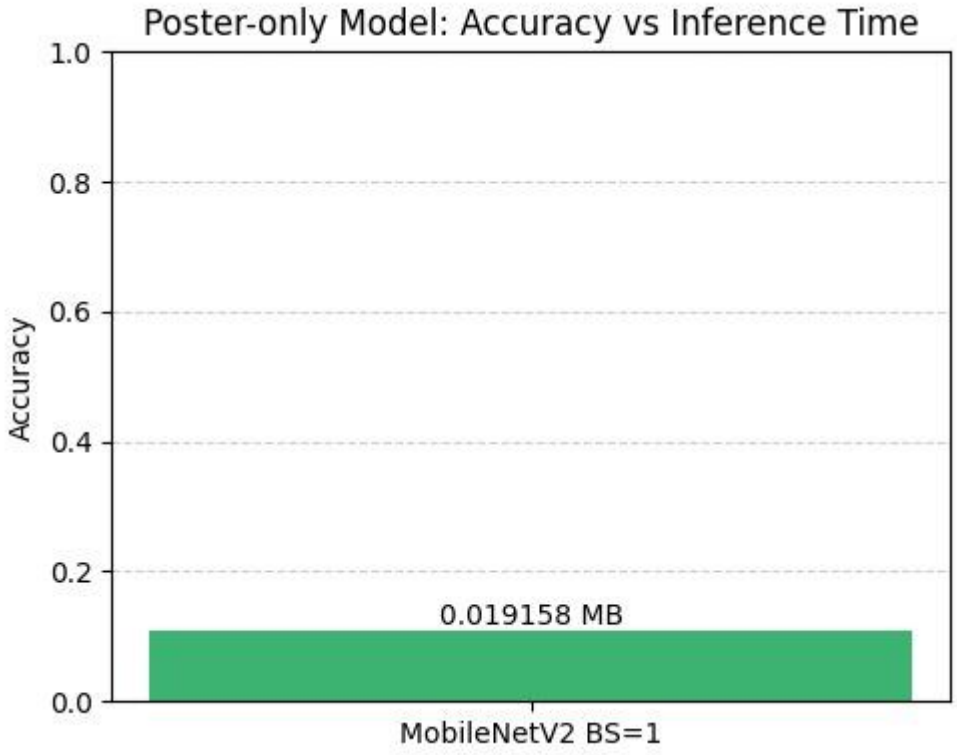
Poster only Unimodel





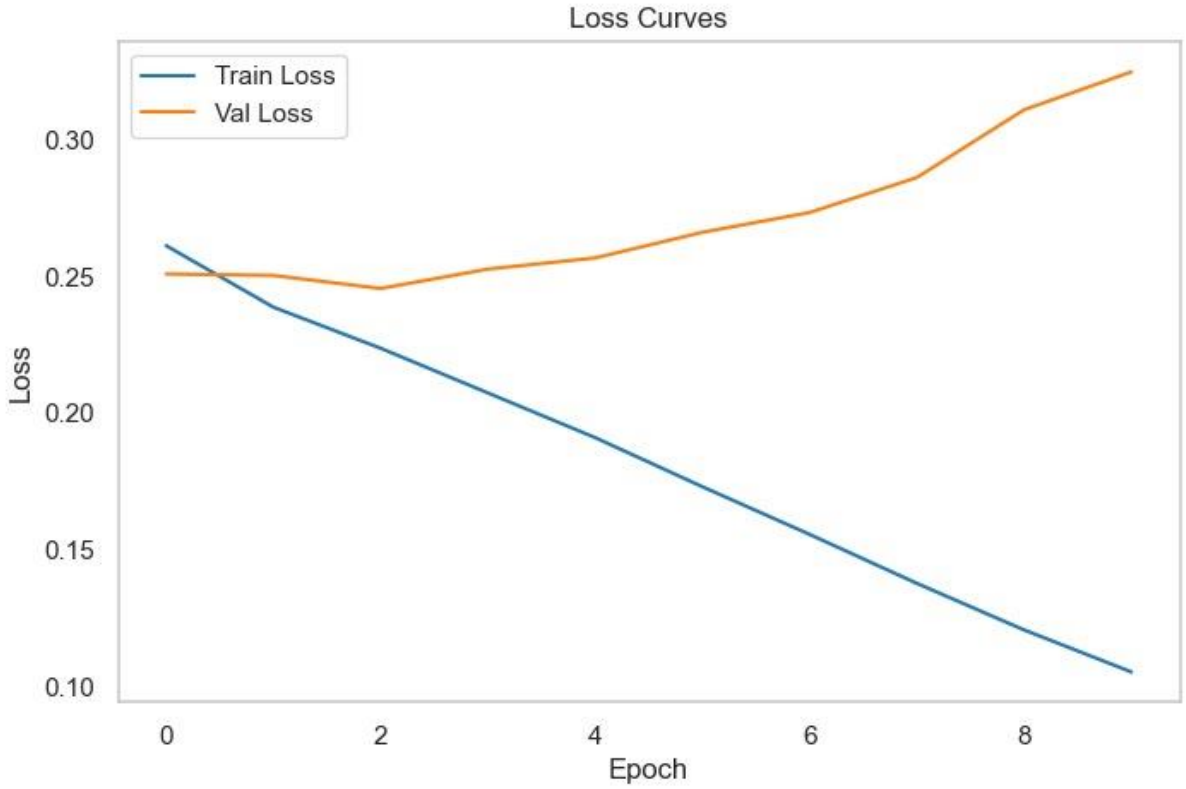

Simplified Multilabel Confusion Matrix
True Positives
False Positives
False Negatives
140 748 80 370 8 45 221 3 189 210 105 171 64 8 19 2 35 39 1 41 43 3 29
95 1567 338 1523 23 396 2105 95 2087 2197 1149 2034 532 175 246 29 991 1686 46 1856 1961 154 2288
1270 107 438 159 386 316 31 273 19 19 84 25 103 127 126 116 65 51 80 14 17 50 3
Action Adventure Animation Biography Comedy Crime Documentary Drama Family Fantasy History Horror Music Musical Mystery News Romance Sci-Fi Short Sport Thriller War Western
2000
1500
1000
500

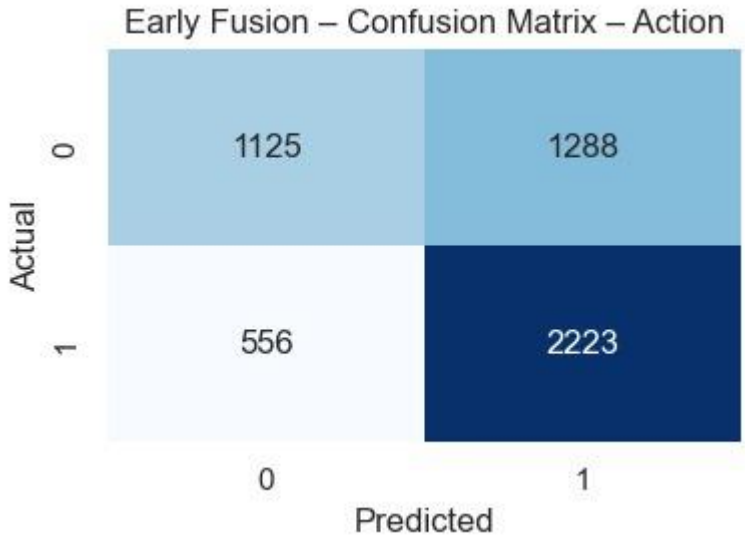

F1 Score per Genre
F1 Score
0.0
0.1
0.2
0.3
0.4
Action Adventure Animation Biography Comedy Crime Documentary Drama Family Fantasy History Horror Music Musical Mystery News Romance Sci-Fi Short Sport Thriller War Western

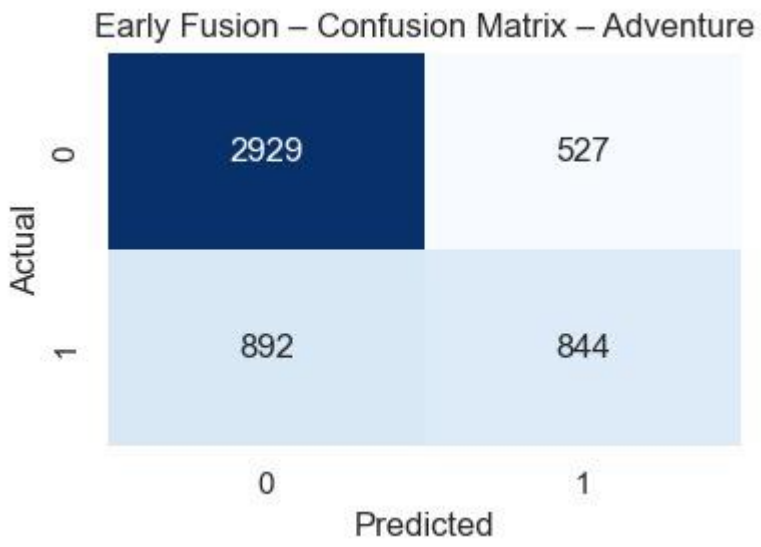

Poster-only Model: Accuracy vs Inference Time
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
0.019158 MB
MobileNetV2 BS=1

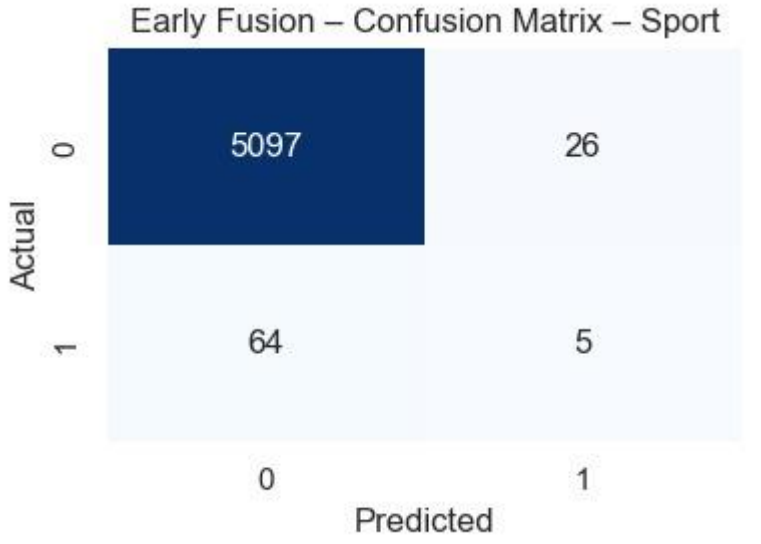

Poster-only Model: F1-score vs Model Size
F1-score
1.0
0.8
0.6
0.4
0.2
0.0
162.03 MB
MobileNetV2 BS=1

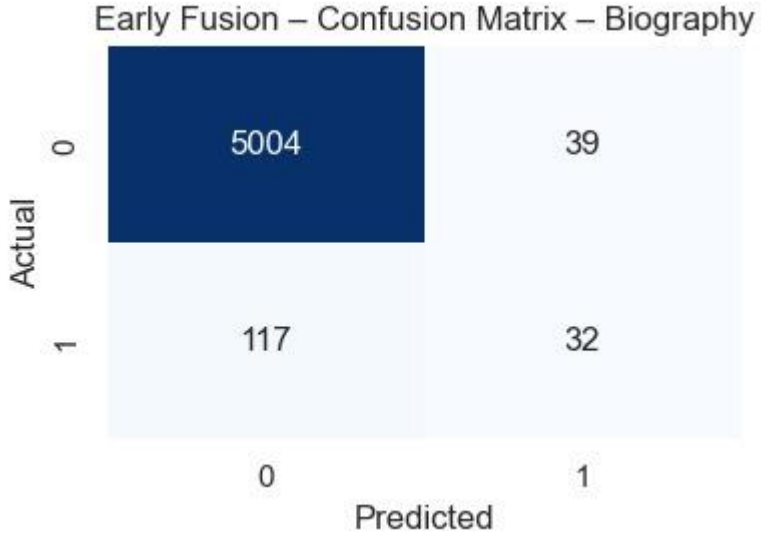
Early fusion (MM-imdb)

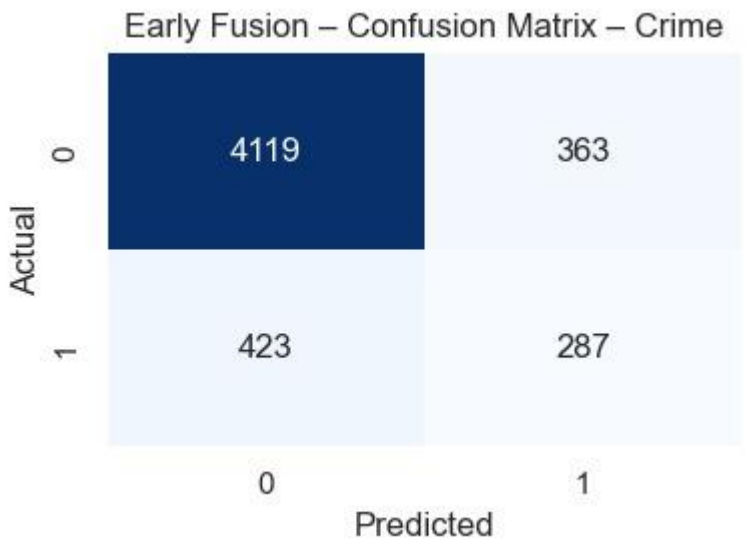


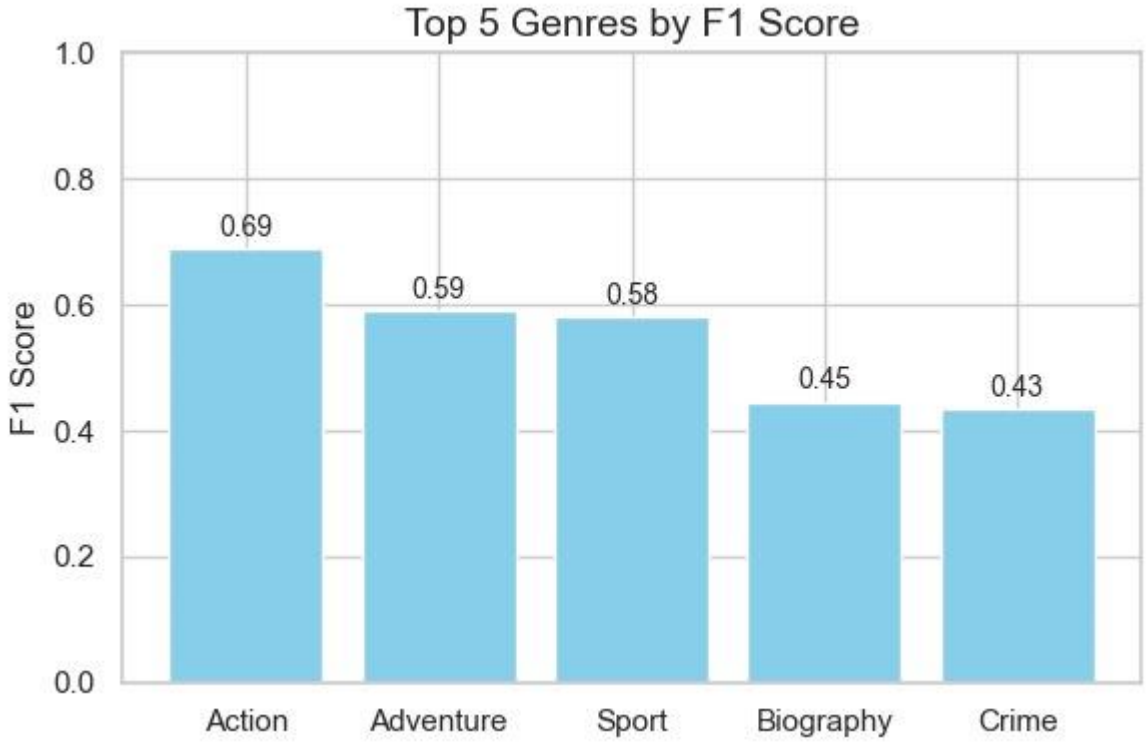


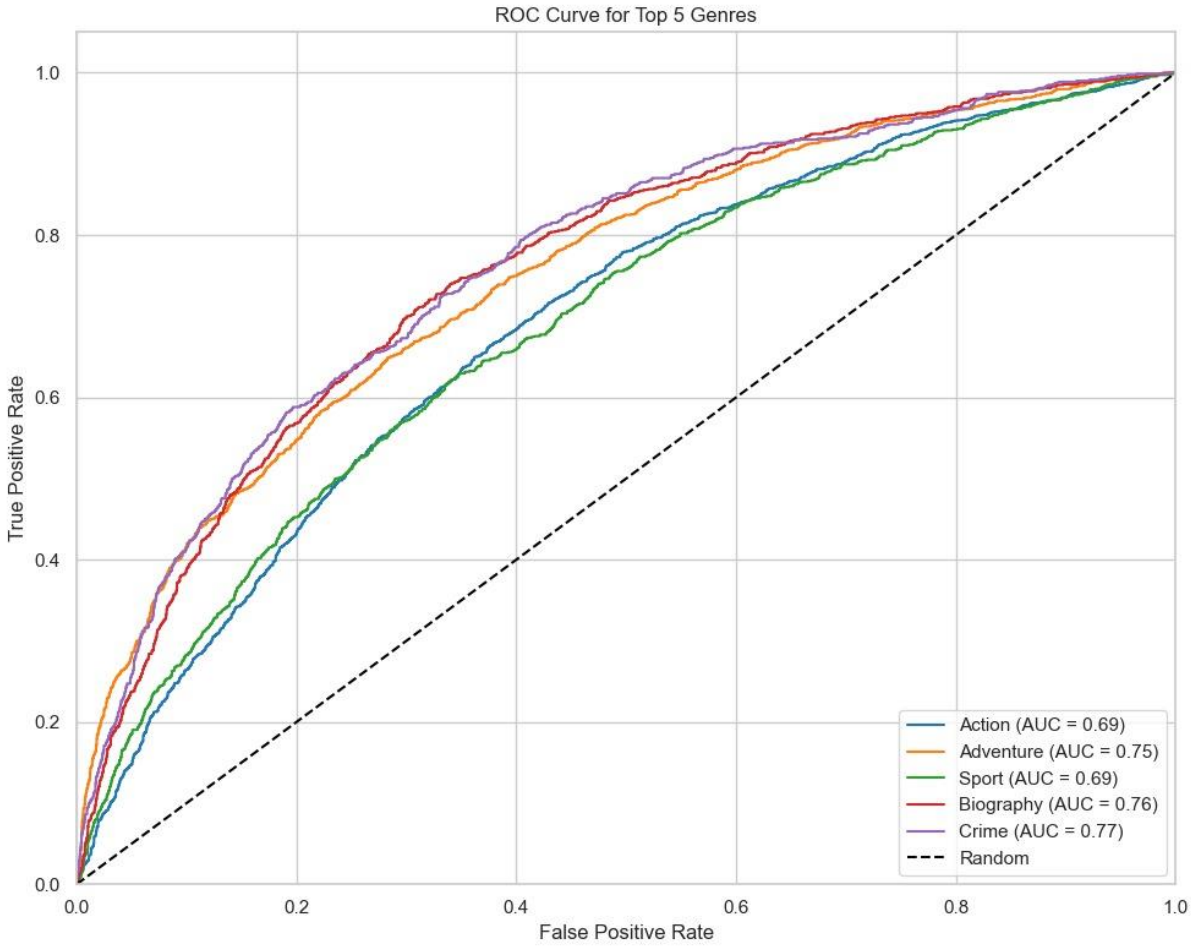

Early Fusion – Confusion Matrix – Adventure
Actual
0
1
2929
527
892
844
0
1
Predicted

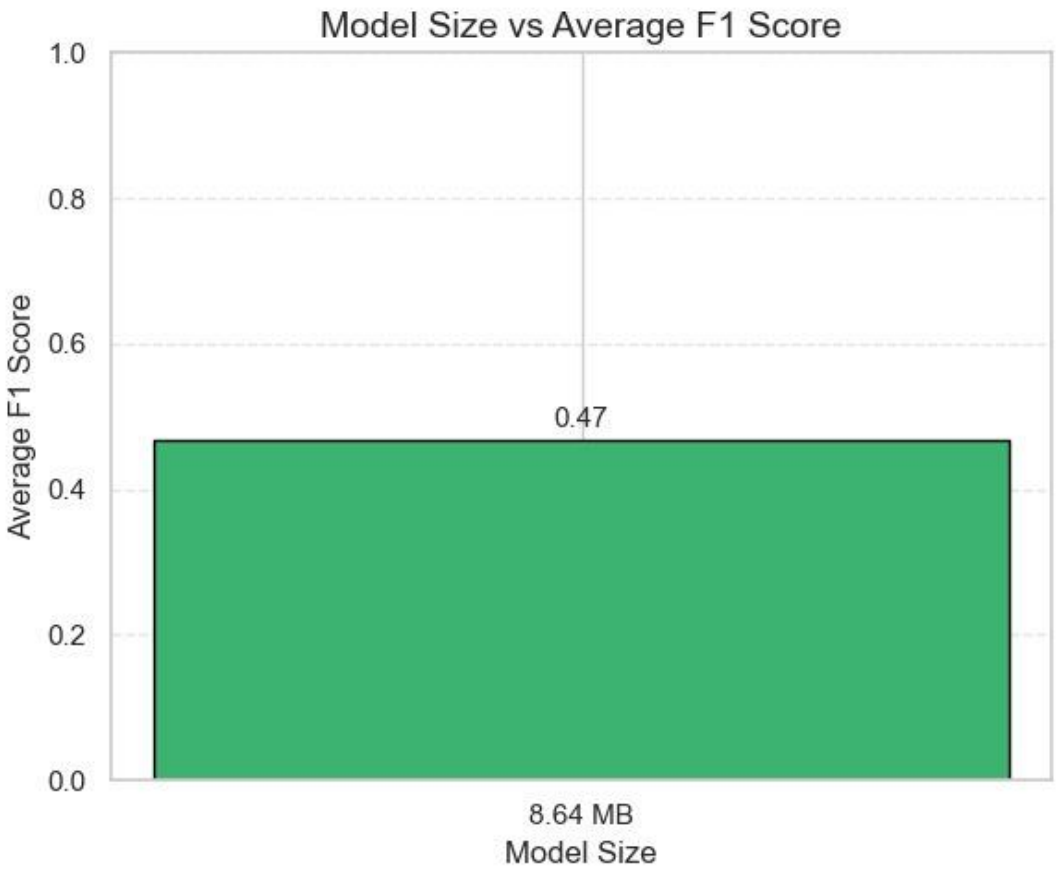

Early Fusion – Confusion Matrix – Sport
Actual
0
1
5097
26
64
5
0
1
Predicted

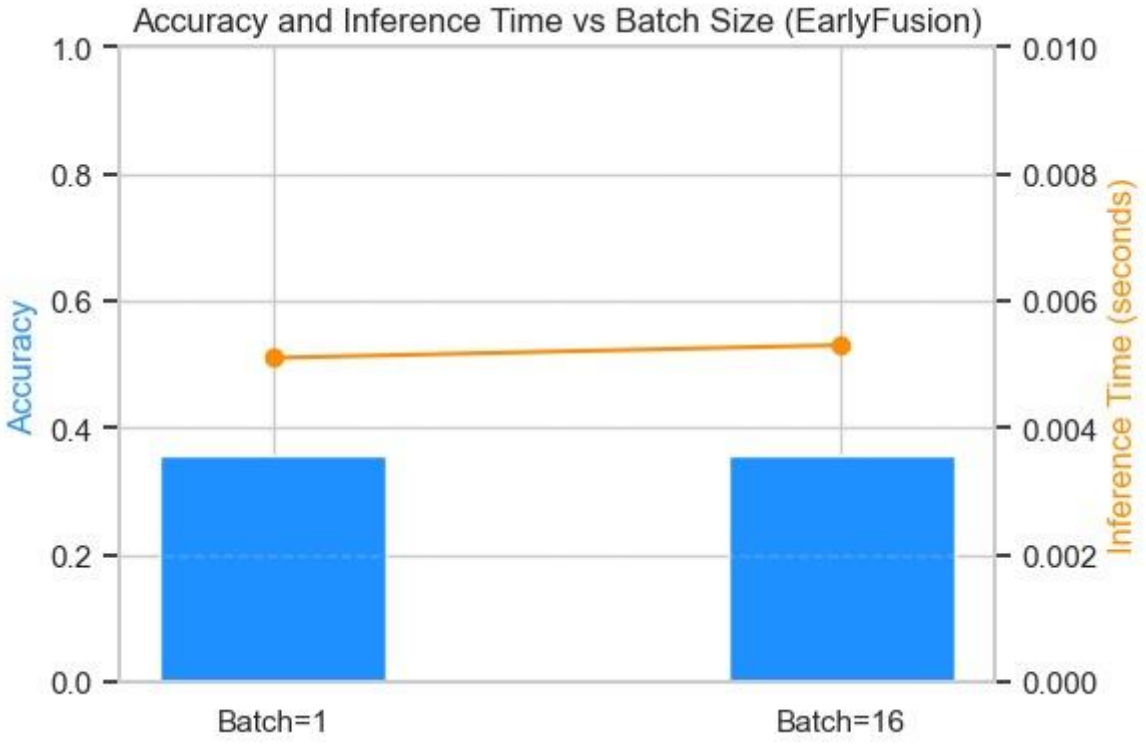

Early Fusion – Confusion Matrix – Biography
Actual
0
1
5004
39
117
32
0
1
Predicted


Early Fusion – Confusion Matrix – Crime
Actual
0
1
4119
363
423
287
0
1
Predicted

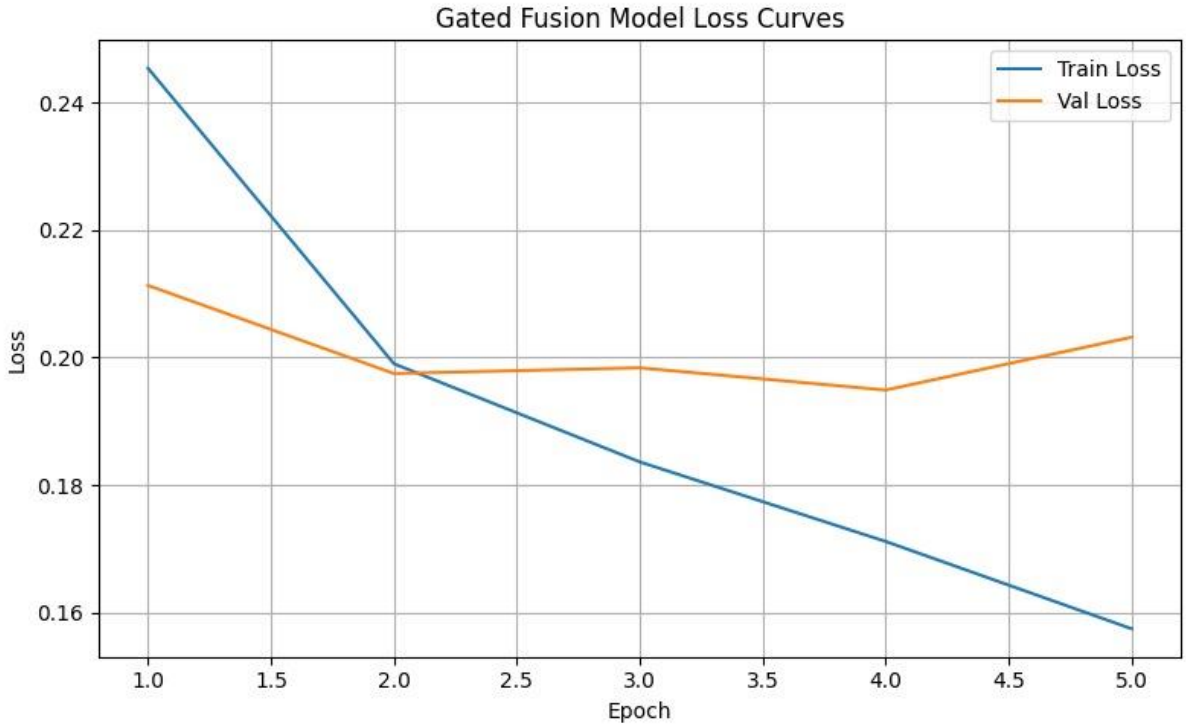

Top 5 Genres by F1 Score
F1 Score
1.0
0.8
0.6
0.4
0.2
0.0
0.69
0.59
0.58
0.45
0.43
Action
Adventure
Sport
Biography
Crime

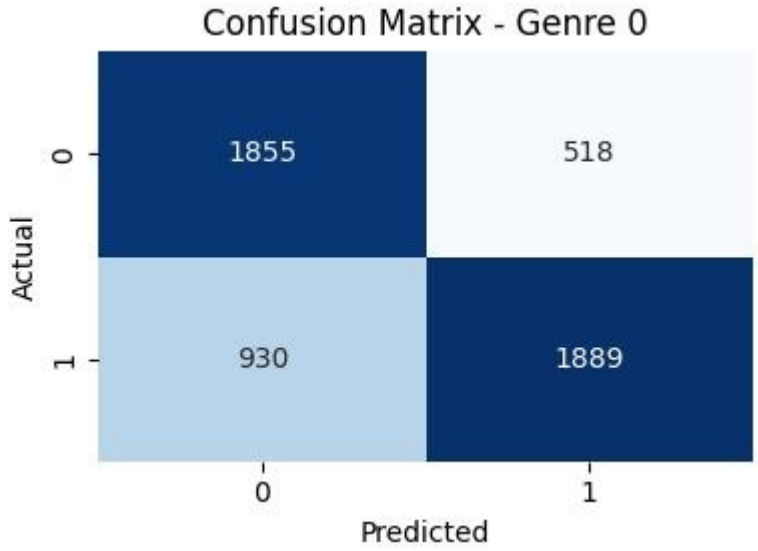

ROC Curve for Top 5 Genres
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Action (AUC = 0.69)
Adventure (AUC = 0.75)
Sport (AUC = 0.69)
Biography (AUC = 0.76)
Crime (AUC = 0.77)
Random

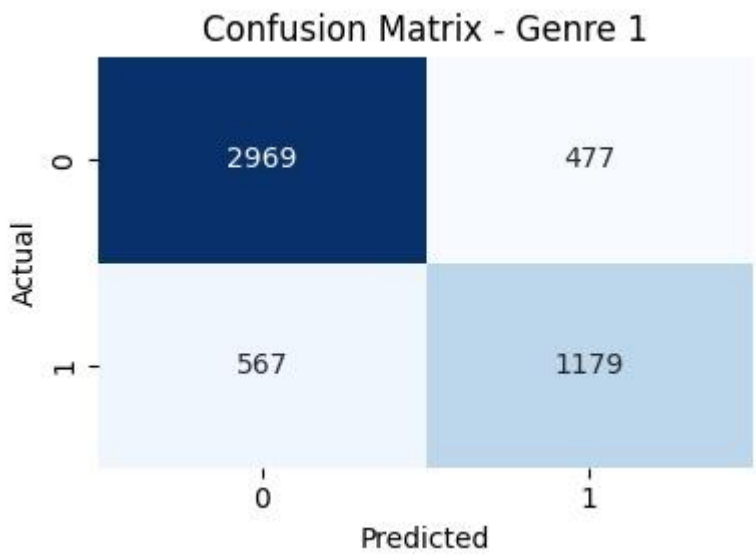

Model Size vs Average F1 Score
Average F1 Score
1.0
0.8
0.6
0.4
0.2
0.0
0.47
8.64 MB
Model Size

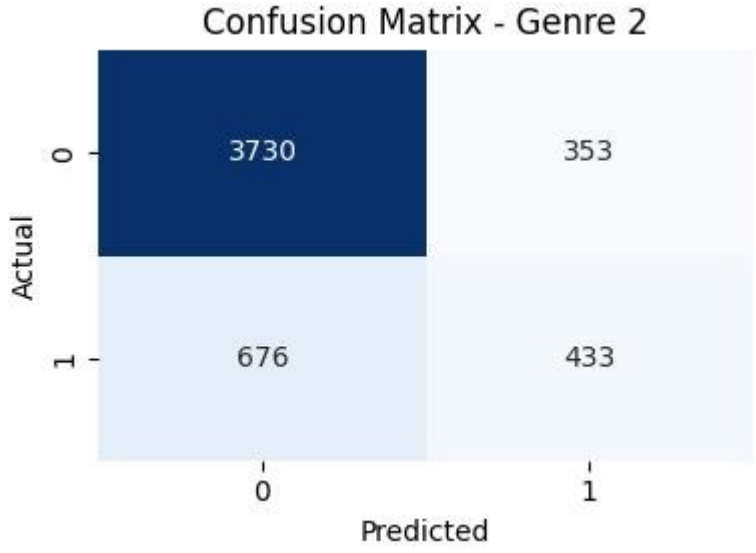

Accuracy and Inference Time vs Batch Size (EarlyFusion)
Accuracy
Inference Time (seconds)
1.0
0.8
0.6
0.4
0.2
0.0
0.010
0.008
0.006
0.004
0.002
0.000
Batch=1
Batch=16

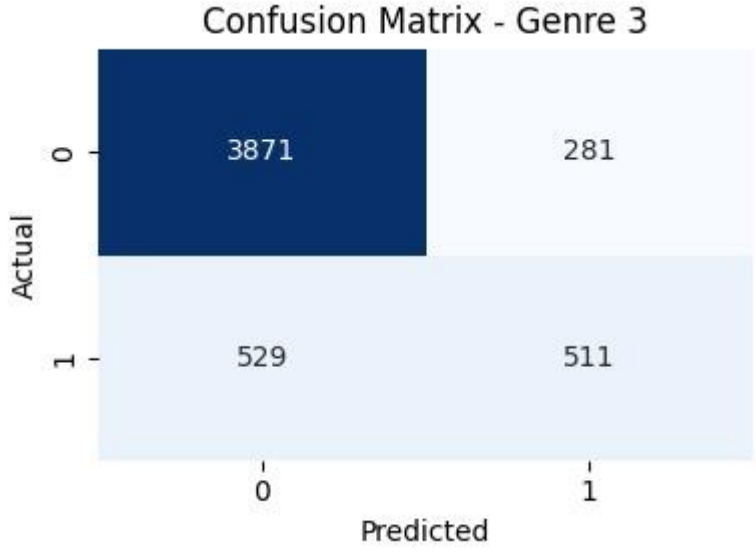
Gated Fusion (MM-imdb)

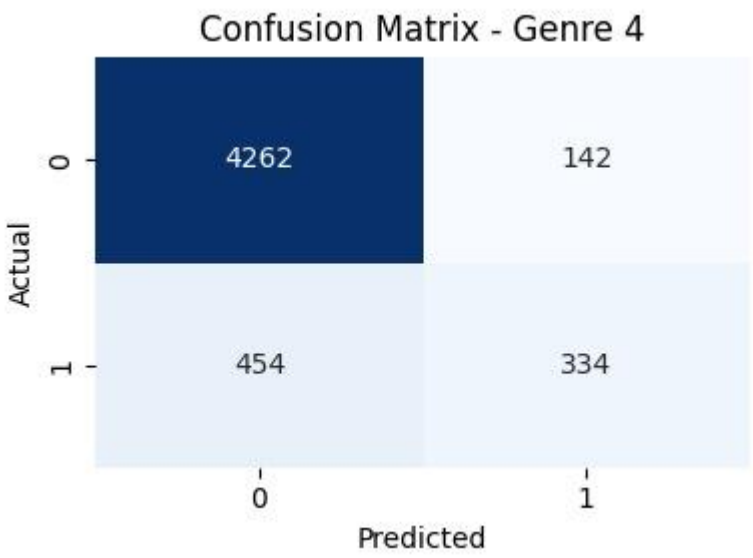


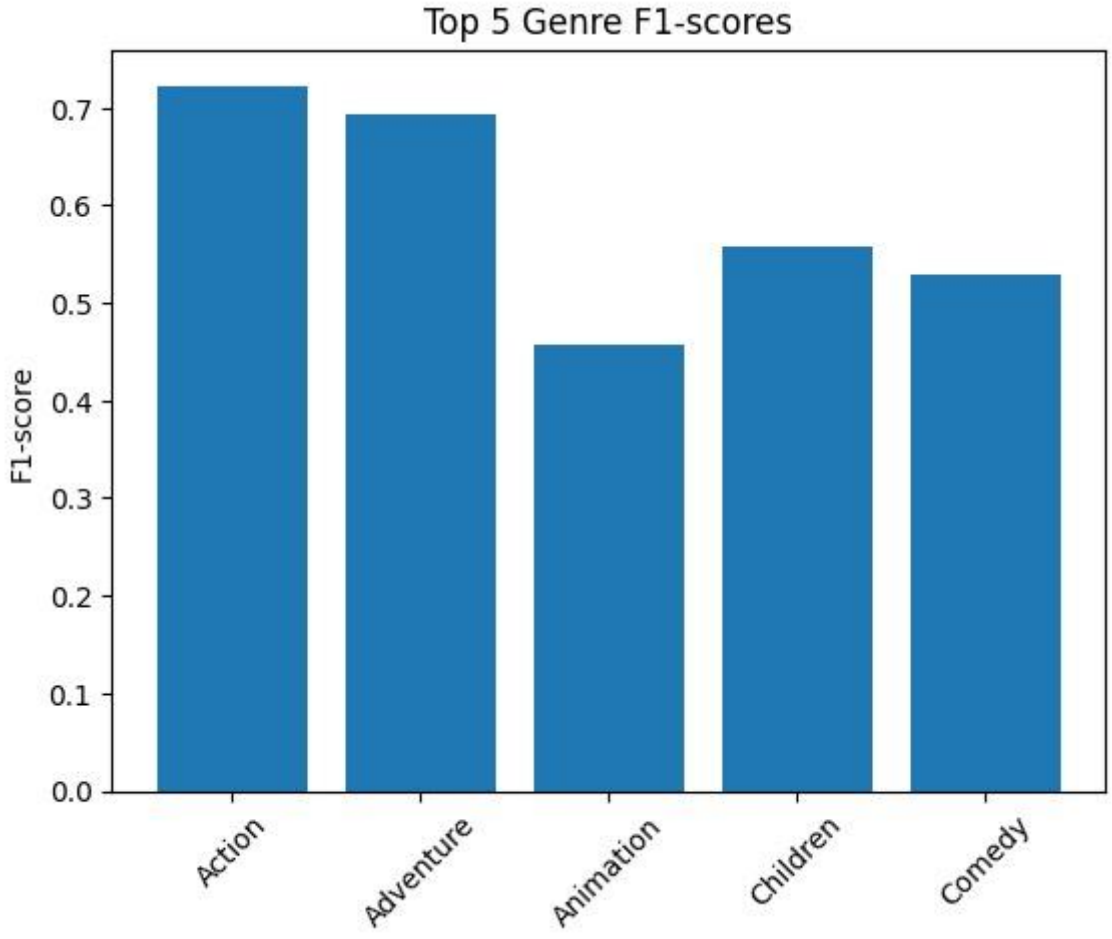


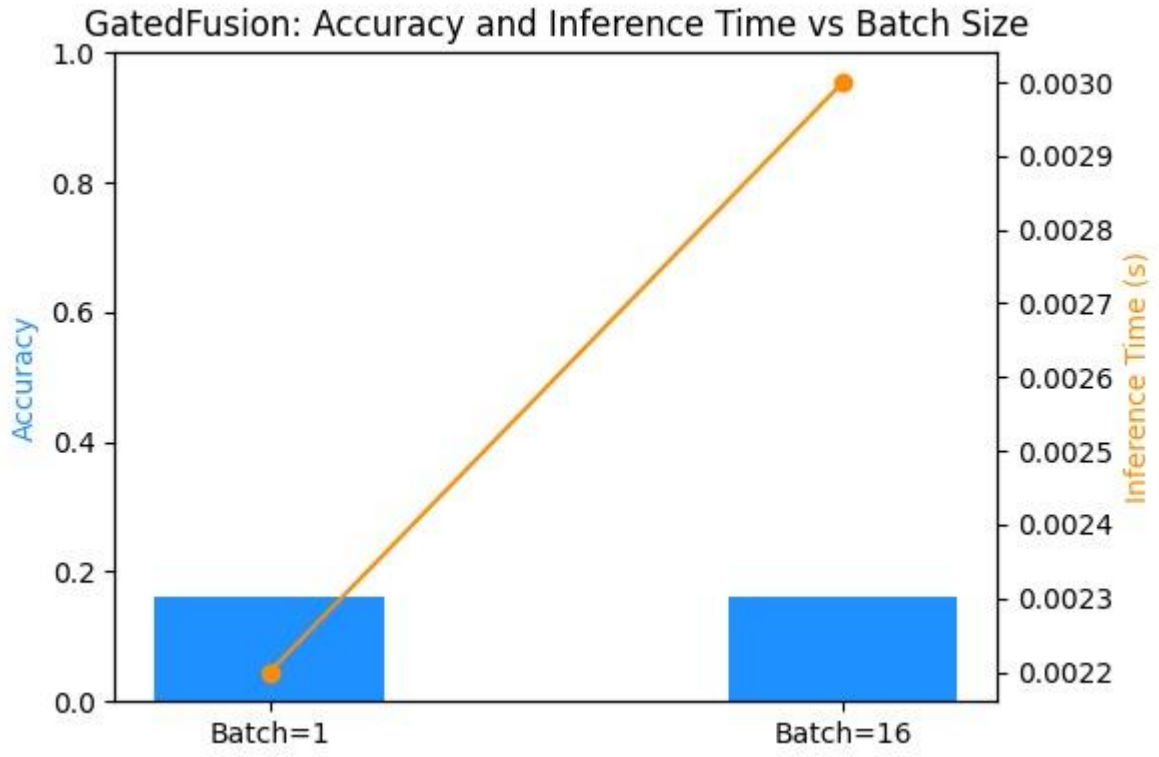

Confusion Matrix - Genre 1
2969
477
567
1179
Actual
0
1
0
1
Predicted

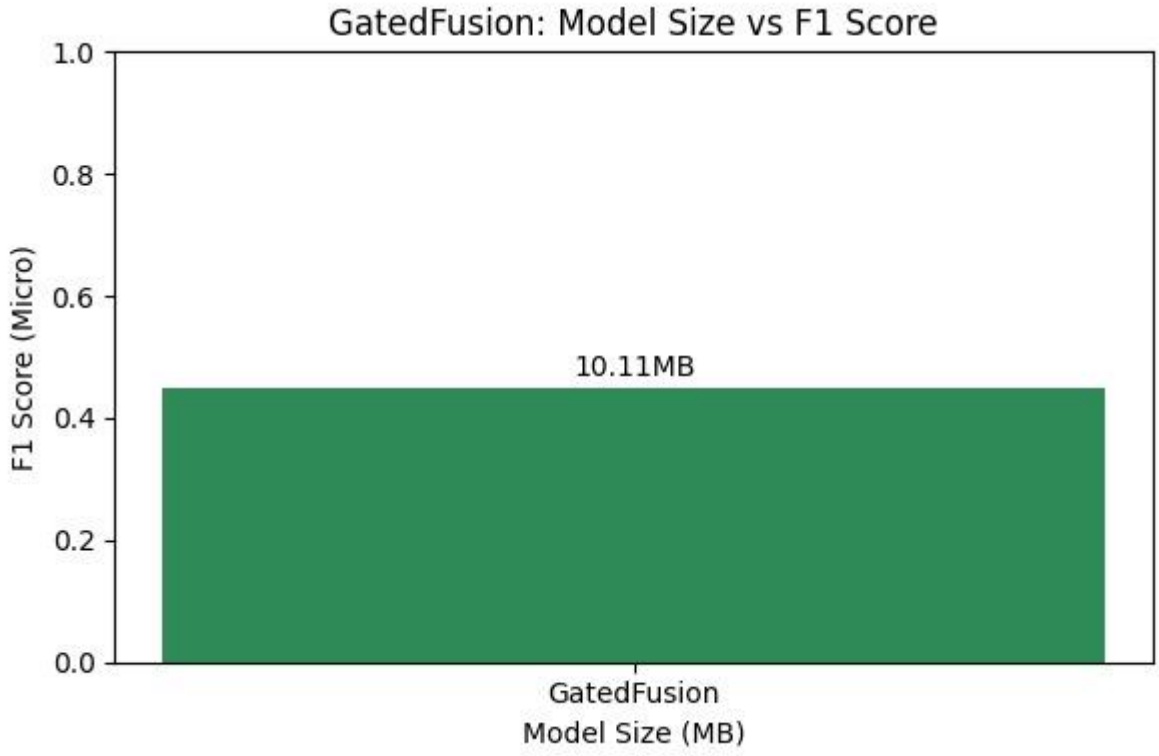

Confusion Matrix - Genre 2
3730
353
676
433
Actual
0
1
0
1
Predicted

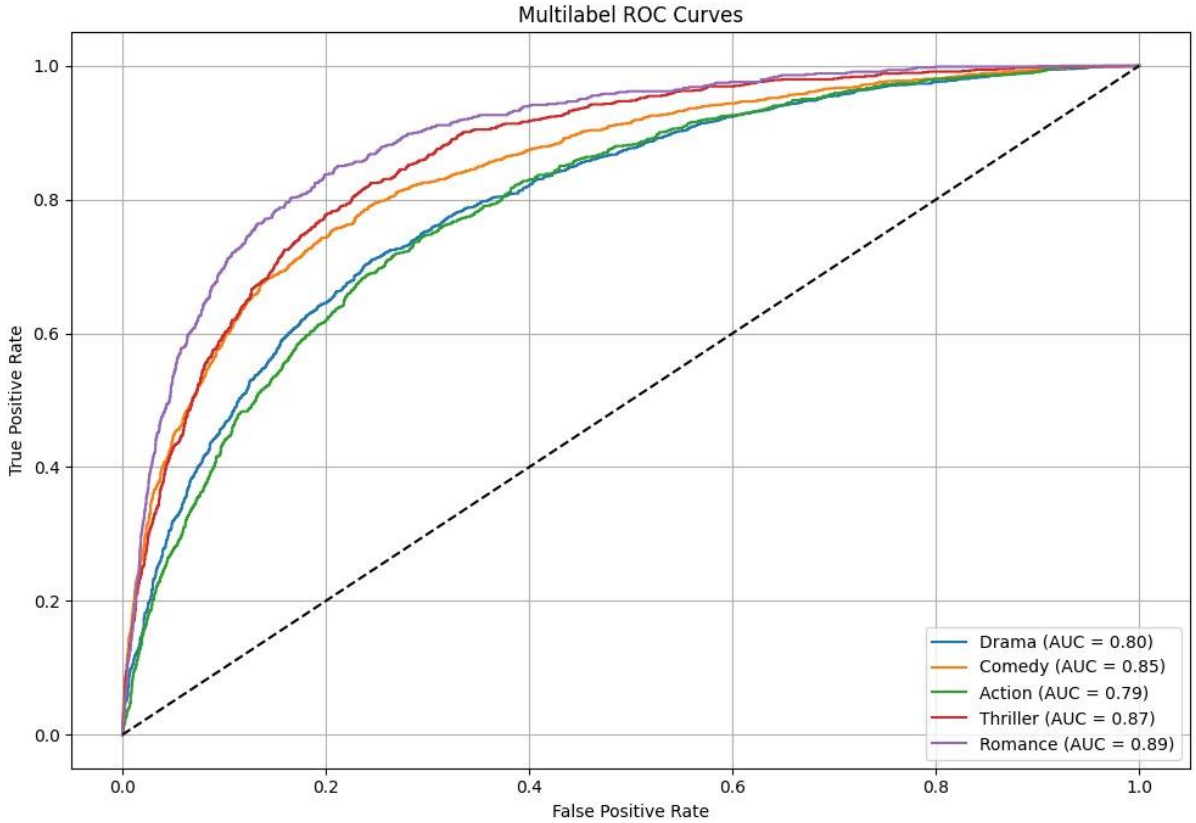

Confusion Matrix - Genre 3
3871
281
529
511
Actual
0
1
0
1
Predicted


Confusion Matrix - Genre 4
4262
142
454
334
Actual
0
1
0
1
Predicted


Top 5 Genre F1-scores
F1-score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Action
Adventure
Animation
Children
Comedy


GatedFusion: Accuracy and Inference Time vs Batch Size
Accuracy
Inference Time (s)
1.0
0.8
0.6
0.4
0.2
0.0
0.0030
0.0029
0.0028
0.0027
0.0026
0.0025
0.0024
0.0023
0.0022
Batch=1
Batch=16


GatedFusion: Model Size vs F1 Score
F1 Score (Micro)
1.0
0.8
0.6
0.4
0.2
0.0
10.11MB
GatedFusion
Model Size (MB)


Multilabel ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Drama (AUC = 0.80)
Comedy (AUC = 0.85)
Action (AUC = 0.79)
Thriller (AUC = 0.87)
Romance (AUC = 0.89)